CHAPTER 8
USER DOCUMENT

8.1 Introduction

The User Document provides clear and structured instructions for interacting with the **Personalized Learning Content Generator (PLCG)** system. The PLCG is designed for students, teachers, parents, trainers, and self-learners who want to generate personalized educational content such as explanations, summaries, examples, MCQs, flashcards, and adaptive quizzes. The system helps users create learning material tailored to the learner's **age, interest, tone preference, and difficulty level**.

8.2 System Access

The PLCG application is deployed using a web-based interface (HTML/JS or Streamlit + FastAPI).
Users can access the system through **any modern web browser** without requiring installation.

The system supports:

- Desktop browsers (Chrome, Firefox, Edge)
- Mobile/tablet browsers (responsive interface)

8.3 User Interface Overview

Upon launching the application, the user is presented with an intuitive and easy-to-navigate interface consisting of the following modules:

• Topic-Based Content Generator

Users enter a topic (e.g., Photosynthesis, Fractions, World War II) and receive:

- Explanation
- Examples
- Concept questions
- Flashcards

Users can specify:

- Learner age
- Interests (sports, gaming, music, etc.)
- Learning tone (concise, conversational, detailed)
- Difficulty level (easy/medium/hard)

The system uses these parameters to generate age-appropriate, engaging educational content.

• Multi-Format Output Section

Displays outputs in neatly separated cards:

- Explanation
- Examples
- Questions (with answers)
- Flashcards
- Raw model output

• Download & Copy Tools

Options for:

- Copying output to clipboard
- Downloading JSON files
- Exporting study material

• Documentation Tab (optional)

Includes:

- System description
- Dataset overview
- Methodology summary
- RAG architecture description

8.4 Steps to Use the System

Step 1: Launch the Application

Open the PLCG web interface hosted online or running locally.

Step 2: Enter Input Details

Fill required fields:

- **Topic –** The subject or concept you want to learn
- **Age –** Helps adjust explanation complexity
- **Interests –** Personalizes examples
- **Tone –** Conversational, concise, playful, detailed

Step 3: Choose Output Types

Select which outputs you want:

- Explanation
- Examples
- Questions
- Flashcards
- Raw JSON response

Step 4: Customize Output Quantity

Specify number of:

- Examples
- Questions
- Flashcards

Step 5: Generate Content

Click the **Generate** button to produce personalized learning material.

Step 6: View and Download Outputs

- Review generated content in clean, card-style sections

- Copy content using “Copy” buttons
- Download JSON using **Download** button
- Re-generate or modify inputs anytime

8.5 Output Interpretation

• Explanation

A simplified or age-appropriate description of the topic.
Used for first-time understanding or quick revision.

• Examples

Real-life and interest-based examples connecting the topic to the learner’s background.

• Questions & Answers

Two conceptual or factual questions with answers created for testing and reinforcement.

• Flashcards

Five short question–answer pairs designed for revision and memorization.

• Raw JSON Response

Structured machine-readable output that represents the model response.
Useful for developers integrating PLCG into other applications.

8.6 User Requirements

Browser Requirements

- Google Chrome
- Mozilla Firefox
- Microsoft Edge
- Safari (mobile/desktop)

System Requirements

- Stable internet connection (for API-based generation)
- JavaScript enabled
- Optional: Python environment (for local users running backend/frontend manually)